

Nikolai Avtsinov

Manual / Automation QA Engineer

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SUMMARY

QA Engineer with 5 years of experience in FPGA verification. I build Python-based test environments, automate result analysis, and identify the root causes of discrepancies in complex systems. I enjoy working with data: sorting and analyzing test results, preparing reports, and visualizing information with charts. Experienced in automating REST API and UI testing.

WHY YOU SHOULD HIRE ME

I consider software testing my long-term career and continue to develop in this field deliberately. My broad technical background helps me quickly understand the architecture of new projects and identify weaknesses that are not visible at the specification level. I pay close attention to requirements, take responsibility for product quality, independently fill knowledge gaps, and see tasks through to completion.

EXPERIENCE

DTC Electronics | QA Engineer

May 2021 – Present

Russia, Voronezh

- Created tests for an undocumented legacy FPGA block whose behavior was unknown to the team and documented its reference behavior. This made it possible to rewrite the block from scratch and verify that the new implementation fully matched the original without regressions.
- Analyzed specifications and requirements for FPGA blocks, prepared test plans and test cases, and applied test design techniques such as boundary value analysis, equivalence partitioning, and combinatorial testing.
- Performed manual and exploratory testing of new blocks before automating the checks.
- Developed a Python-based test environment for verifying Verilog FPGA models, including operating modes, switching logic, core functionality, and individual blocks. Separated test infrastructure from test cases, allowing new tests to be added without modifying the framework.
- Automated simulation log analysis with bash scripts, including parsing, filtering, and summarizing run results. Reduced manual analysis effort by 75%.
- Identified and documented more than 2,000 discrepancies between the model and the specification before the synthesis stage. Created bug reports in Jira, investigated defect root causes, and tracked them through resolution.

PROJECTS

Parabank: UI AND REST API TEST AUTOMATION | PYTEST, SELENIUM | 2023

A test suite for the Parabank demo banking application. Covered both the REST API and the web interface. Used pytest as the test framework, Requests for HTTP requests, Pydantic for data model validation, and Selenium WebDriver for UI testing. Set up CI with GitHub Actions and Allure Report for test reporting.

SKILLS

PROGRAMMING

Python • Bash

LIBRARIES / FRAMEWORKS

pytest • Allure • Requests • Selenium • Pydantic

DOCUMENTATION

LaTeX • Markdown • Notion • Confluence

TOOLS / PLATFORMS

Subversion • Git • Postman • GitHub Actions • Docker • Jira • Codex • Cursor • Claude Code

FPGA

Verilog • SKILL (Cadence) • NC-Verilog

EDUCATION

VSTU

BACHELOR'S DEGREE IN ELECTRONICS AND NANO-ELECTRONICS

2015 – 2019 | Voronezh, Russia

VSTU

MASTER'S DEGREE IN ELECTRONICS AND NANOELECTRONICS

2019 – 2021 | Voronezh, Russia

LANGUAGES

RUSSIAN

native

ENGLISH

written B1
spoken A2